

Left Shift : $\text{number} \times 2^n$

Right Shift : $\frac{\text{number}}{2^n}$

SHL/SHR

SAL/SAR

ROL/ROR

RCL/RCR

operation $\text{seg/mem8}, \text{imm8/cl}$

SHLD/SHRD operation $\text{seg16/32/mem16/32}, \text{seg16/32}, \text{imm8/cl}$

Shifting Words:-

SHL BYTE PTR v1, 1

84	12
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SHL RCL BYTE PTR v1+1, 1

Shifting double Words:-

SHL BYTE PTR v1, 1

84	84	84	84
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RCL BYTE PTR v1+1, 1

// // v1+2, 1

// // v1+3, 1

Multiplication & Division

Unsigned

Signed

→ MUL $\text{seg}/\text{mem}8$
 $\text{seg}/\text{mem}16$
 $\text{seg}/\text{mem}32$

→ IMUL $\text{seg}/\text{mem}8$
 $\text{seg}/\text{mem}16$
 $\text{seg}/\text{mem}32$

multiplicand	multiplier	Product
AL	$\text{seg}/\text{mem}8$	AX
AX	$\text{seg}/\text{mem}16$	DX:AX
EAX	$\text{seg}/\text{mem}32$	EDX:EAX

→ in IMUL, the 2-operand format causes the product to fit the destination. CF/OF is set when digits are lost.

CF=1 if
DX $\neq$ 0

IMUL $\text{seg}16, \text{seg}/\text{mem}16 / \text{imm}8/16$

IMUL $\text{seg}32, \text{seg}/\text{mem}32 / \text{imm}8/32$

→ the 3-operand format stores result in first op, when multiplies 2nd & 3rd op. CF/OF = 0 if sign extension in upper byte.

IMUL $\text{seg}16, \text{seg}/\text{mem}16, \text{imm}8/16$

IMUL $\text{seg}32, \text{seg}/\text{mem}32, \text{imm}8/32$

Division

Dividend	Divisor	Quotient	Remainder
AX	seg/mem8	AL	AH
DX: AX	seg/mem16	AX	DX
EDX: EAX	seg/mem32	EAX	EDX

→ in signed division, the dividend must be sign extended.

CBW ⇒ extends sign bit AL to AH
 CWD ⇒ // Ax to DX
 CDQ ⇒ // EAX to EDX